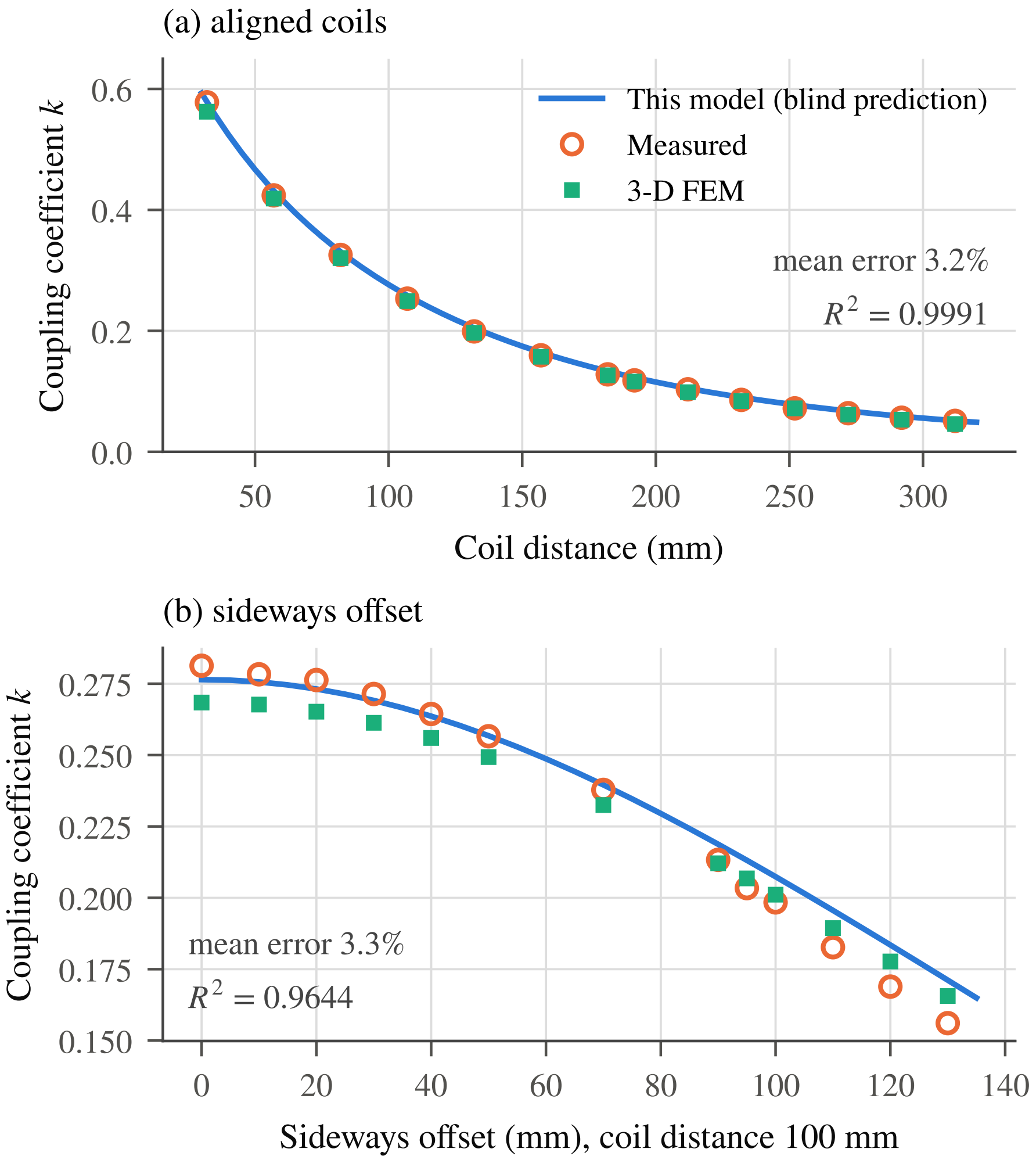
(a) aligned coils
Coupling coefficient k
0.6
0.4
0.2
0.0
50
100
150
200
250
300
Coil distance (mm)
This model (blind prediction)
Measured
3-D FEM
mean error 3.2%
$R^2 = 0.9991$
(b) sideways offset
Coupling coefficient k
0.275
0.250
0.225
0.200
0.175
0.150
0
20
40
60
80
100
120
140
Sideways offset (mm), coil distance 100 mm
mean error 3.3%
$R^2 = 0.9644$